

Tropical Cyclone Genesis — Part I

Purpose. Use this sheet to record the main conceptual steps in tropical cyclogenesis and to distinguish environmental favorability from an actual genesis mechanism. The activities follow the sequence of the Part I lecture.

Learning goals

By the end of this activity, you should be able to:

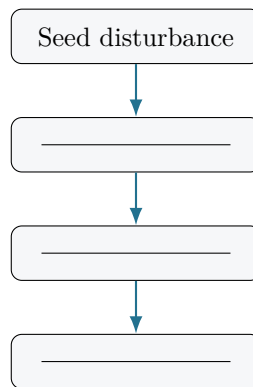
- distinguish **favorable conditions** from a **genesis mechanism**;
- explain why TC genesis is a **finite-amplitude, multiscale transition**;
- compare the conceptual **top-down** and **bottom-up** views;
- explain the Bister–Emanuel (1997) pathway from an MCS to a surface cyclone;
- describe how column moistening, vorticity stretching, boundary-layer inflow, and surface enthalpy flux interact during genesis.

Activity 1 — What do we mean by “genesis”?

(8 min)

1.1 Build the conceptual sequence

Complete the progression discussed in lecture.



1.2 Operational versus dynamical genesis

In one or two phrases, record what each viewpoint emphasizes.

Operational view:

Dynamical view:

1.3 Why is genesis difficult?

The lecture describes TC genesis as both an **upscale-organization problem** and a **finite-amplitude threshold problem**. Explain each phrase in your own words.

Upscale organization:

Finite-amplitude threshold:

Activity 2 — Conditions are not mechanisms**(8 min)****2.1 Sort the ingredients**

For each item below, decide whether it is primarily an **environmental/favorable condition** or a **process within a genesis mechanism**. Some items may interact with both, but choose the role emphasized in the lecture.

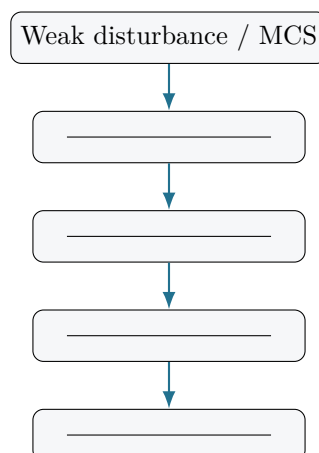
Ingredient/process	Condition	Mechanism
Weak vertical wind shear	☐	☐
High midtropospheric humidity	☐	☐
Pre-existing low-level disturbance	☐	☐
Vorticity stretching	☐	☐
Boundary-layer spinup	☐	☐
Warm ocean / large air–sea disequilibrium	☐	☐
Vortex aggregation	☐	☐
Persistent convective organization	☐	☐

2.2 The key distinction

Why can a disturbance exist in an apparently favorable tropical environment and still fail to become a tropical cyclone?

Activity 3 — Bister & Emanuel (1997): the “top-down” pathway**(15 min)****3.1 Conceptual chain**

Complete the sequence.



3.2 How does the midlevel vortex form?

Use the lecture's description of stratiform precipitation, evaporation/melting, nonuniform diabatic heating, and convergence to explain the process.

The lecture used the approximate stretching relation

$$\frac{D\zeta_a}{Dt} \approx -\zeta_a \nabla_h \cdot \mathbf{v}, \quad \zeta_a = f + \zeta.$$

For cyclonic absolute vorticity ($\zeta_a > 0$), what happens to ζ_a when horizontal convergence occurs ($\nabla_h \cdot \mathbf{v} < 0$)?

4.3 Why is the initial vortex elevated and cold-core?

Give the two reasons emphasized in the lecture.

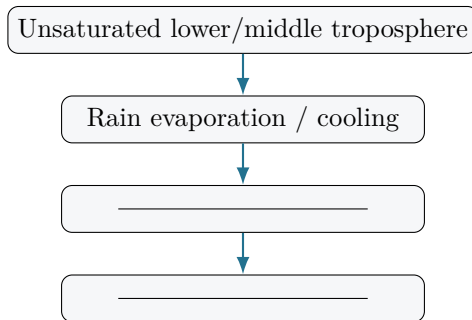
1. _____
2. _____

4.4 Does the vortex literally “descend”?

The lecture explicitly states that the midlevel vortex does *not* simply sink as a rigid object. Describe what actually develops downward toward the surface.

Activity 4 — Thermodynamic preconditioning**(7 min)****4.1 The dry-air penalty**

Complete the cause-and-effect chain.

**4.2 Why does moistening matter?**

Explain why repeated rain and convection can make later convection more effective at spinning up a low-level cyclone.

Activity 5 — Surface enthalpy flux and WISHE**(10 min)**

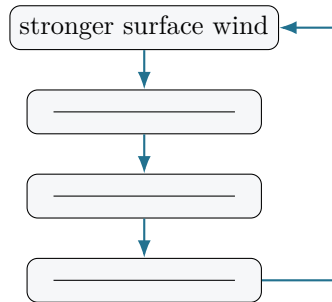
The lecture writes the surface enthalpy flux approximately as

$$F_k \approx \rho C_k |V| (k_s^* - k_a).$$

5.1 Define the terms

F_k	_____
C_k	_____
$ V $	_____
k_s^*	_____
k_a	_____

5.2 Complete the feedback loop



5.3 Short calculation / interpretation

Suppose ρ , C_k , and $(k_s^* - k_a)$ remain approximately fixed while the surface wind speed doubles. According to the bulk-flux expression, how does F_k change? What physical feedback might follow?

5.4 Why preconditioning first?

Why is surface-flux feedback more effective *after* the column has become moist and downdrafts have weakened?

Activity 6 — Put the theory together**(5 min)**

In 4–6 sentences, explain the Bister–Emanuel genesis pathway to a classmate. Your explanation must use all of the following terms correctly:

**stratiform precipitation midlevel vortex column moistening boundary-layer inflow
surface enthalpy flux**

Reflection**(3 min)**

1. What is the most important distinction between a favorable genesis environment and a genesis mechanism?

2. In the Bister–Emanuel picture, what is the key transition that allows an elevated midlevel disturbance to become a surface-based tropical cyclone?

3. Write one question you still have about tropical cyclogenesis.